(21213) Roll No. .. ..

BCA-V Sem.

18021

B. C. A. Examination, Dec. 2013

Introduction to DBMS

(BCA-501)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt all the Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. Explain instances and schemas in brief.

2. Define inheritance in terms of DBMS.

3. Explain the three level architecture of database management system in brief.

4. What are the differences between logical data independence and physical data independence ?

5. What is meant by data model ? Explain Relational model.

Section-B

(Short Answer Questions)

Answer any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 7½×2=15

6. Construct an E-R diagram that consists of employee, job and branch related through the relationship set works on.

7. What are weak entity sets ? Explain the primary key, secondary key and candidate key.

8. Explain the term generalization and specialization.

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. What is structured query language ? Give its characteristics and advantages.

10. Consider the following relational schema :

(An employee can work in more than one department)

Emp (Eid, Ename, Salary)

Works(Eid, did)

Dept (did, dname, managerid, floornum)

Write the following in SQL

(i) Print the names of the departments that employees who works on floor(s) where Mukesh Rawat works.

(ii) Print the names of all employees who works on floor (s) where Mukesh Rawat works.

11. What is normalization ? Explain in detail 1st, 2nd and 3rd Normal Form in detail with importance.

12. What do you mean by hashing and explain all collision resolution techniques in detail ?

13. What do you understand by transaction system ? Give the ACID properties of transaction system.

M

(Printed Pages 4)

(20115)

Roll No. .....

BCA - V Sem.

18021

BCA Examination, Dec. 2014

Introduction to DBMS

(BCA-501)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note: Attempt **all** the Sections as per instructions.

Section-A

Note: Attempt all the **five** questions. Each question carries **03** marks. 3×5=15

1. What do you mean by Database? What are the characteristics of Database approach.

2. State the difference between Generalization & Specialization.

P.T.O.

3. Differentiate between Sequential and Random access in files.

4. What do you mean by relational algebra? What are the various relational Algebra operators?

5. Define 3rd NF and BCNF with example.

Section-B

Note: Attempt any **two** questions out of the following 03 questions. Each question carries 7.5 marks. 7.5×2=15

6. What do you mean by Integrity? Explain various Integrity constraints with example.

7. What are the various database design guidelines? Explain 4th NF and 5 NF.

8. Create ER diagram for the following relations:

Emp

Empno	E name	Sal Joingdate deptno

Dept

Deptno	Dname	Loc

Project

Projno	Deptno	Projname	No. of projects

Section-C

Note: Attempt any **three** questions out of the following **5** questions. Each question carries **15** marks. 3×15=45

9. Explain DBMS architecture in detail.

10. What do you mean by data Independence? What is the difference between logical data Independence and physical data Independence? https://www.ccsustudy.com

11. Define Entity, Entity sets, strong entity, weak entity and Draw ER diagram for School management system by taking your own assumptions.

12. What do you mean by Normalization? Why do we normalize the database?

13. Differentiate between Association and Aggragation with suitable examples.

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(20116)

Roll No. .....

BCA - V Sem.

18021

BCA Examination, Dec. 2015

Introduction to DBMS

(BCA-501)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section - A

Note : Attempt all the **five** questions. Each question carries **03** marks.

1. What is a data model? Why is it important?

2. What is the difference between entity, entity type and entity sets?

3. How do we convert a normalized relation to that in 1NF? Explain with example.

P.T.O.

4. What do you understand by hashing? Explain with example.

5. What are nested SQL queries? Explain with example.

Section - B

Note : Attempt any **two** questions out of the following **03** questions. Each question carries **7.5** marks.

6. Explain the concept of functional dependency and full functional dependency with an example. Also explains 2NF in detail.

7. Explain the concept of keys. Also explain primary, candidate, super and foreign key with suitable example.

8. What do you mean by data independence? Explain the difference between logical and physical data independence.

Section - C

Note : Attempt any **three** questions out of the following **05** questions. Each question carries **15** marks.

9. Explain the three level architecture of DBMS. Also explain objective of using three level architecture of DBMS.

10. How is ER model mapped to relational model? Explain in detail.

11. What is a lock? Explain in detail about Binary, Shared and Exclusive locks.

12. Define normalization. Why do we need to normalize the database? Also explain 1NF, 2NF and 3NF in detail.

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(21216) Roll No. .....

BCA- V Sem.

18021

B. C. A. Examination, Dec. 2016

Introduction to DBMS

(BCA-501)

(New Course)

Time : Three Hours] *[Maximum Marks : 75*

Note: Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Answer all the *five* questions. Each question carries 3 marks. Very short answer is required. 3×5=15

1. Explain the various categories of data model.

2. What is the difference between subclass and super-class in object model?

3. What are the anomalies in 1NF?

4. Explain in detail about indexed sequential access file.

5. What are the different relational algebraic constraints?

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Section-B

(Short Answer Questions)

Answer any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. How are the anomalies of 1NF are rectified? Explain with the help of example.

7. What are integrity rules? Explain with example.

8. What do you understand by mapping? Explain the different types of mapping.

Section-C

(Detailed Answer Questions)

Answer any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. What is the difference between file system approach and database approach? Also explain component of database system.

10. How are EER diagrams for specialization, generalization and aggregation drawn?

11. Explain the ACID property of a transaction and the usefulness of each with example.

12. Define normalization. Why do we need to normalize the database? Also differentiate 3NF and BCNF in detail.

13. What do you understand by E-R diagrams? Draw the E-R diagram for college management system with E-R functionalities.

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(201217) Roll No. .....

BCA- V Sem.

18021

BCA Examination, Dec- 2017

Introduction to DBMS

(BCA-501)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** questions as per the Instructions.

Section-A

Note : Attempt all **five** questions. Each question carries **three** marks. 5×3=15

1. Differentiate between DDL and DML.
2. List three main advantages of database approach.
3. What is relational algebra? Explain.
4. Briefly describe B tree.
5. Explain ER diagram.

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Section-B

Note : Attempt any **two** questions out of following 3 questions. Each question carries equal marks. 2×7.5=15

6. Discuss the three level architecture of DBMS.
7. Explain different collision resolution approaches in hashing.
8. What is strong and weak entity? Explain with example.

Section-C

Note : Attempt any **three** questions out of following 5 questions. Each question carries equal marks. 3×15=45

9. With respect to the relational data model, define and relate with example: primary key, foreign key, super key, candidate key and prime attribute.
10. Consider the following relations with underlined primary keys.

    PRODUCT (Pcode, Description, Stocking Date, QtyOnHand, MinQty, Price, Discount, VCode)

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VENDOR (VCode, Name, Address, Phone)

Here a vendor can supply more than one product but a product is supplied by only one vendor.

Write SQL queries for the following:

(i) List the names of all the vendors who supply more than one product.

(ii) List the details of the products whose prices exceed the average product price.

(iii) List the Name, Address and Phone of the vendors who are currently not supplying any product.

11. List advantages and disadvantages of Indexed sequential, B tree and B+ tree file organization.

12. Write short notes on:

(i) Functional dependency

(ii) Types of SQL queries

(iii) Transaction processing

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13. Given the following tables:

DIRECTOR

DIR_NUM	DIR_NAME	DIR_DOB
100	Arvind Gaur	30/6/43
101	Faizal Alkazi	12/8/50
102	Anuradha Kapoor	21/9/62

PLAY

PLAY_CODE	PLAY-NAME	DIR-NUM
1001	Jivit ya Mrit	102
1002	Bade Bhai Saab	101
1003	Galib in Delhi	102
1004	Safarnama	100
1005	Aadhe Adhure	101
1006	Konark	101
1007	Adhoori Kahani	NULL

(i) Identify the Primary and Foreign keys of both the tables.

(ii) Do the tables exhibit Entity and Referential Integrity Constraints? Explain.

(iii) Draw the entity relationship diagram to show the relationship between director and play.

Specify the cardinality and participation constraints also.

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(21218)

BCA-V Sem.

Roll No. .....

18021

B. C. A. Examination, Dec. 2018

Introduction to DBMS

(BCA-501)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required.

3×5=15

1. Define super keys, candidate keys and primary keys.

2. Explain the term – data independence.

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3. What is hashing ? Explain.

4. What do you mean by relational database ? Describe.

5. Briefly explain the terms – security and authorization.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. What is a referential integrity constraint ? Why do we need it ? Explain with the help of suitable example.

7. List four significant differences between a file-processing system and a DBMS.

8. Explain third normal form with example.

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Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. Explain the different locking techniques with example for concurrency control.

10. Explain indexed sequential, B tree and B+ tree file organization with example.

11. Explain super class, inheritance and generalization with example.

12. Write short notes on the following :
    (i) Strong and weak entity
    (ii) Different levels of data abstraction
    (iii) Database recovery techniques.

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13. A hospital maintains data about the following entities :
    (i) PATIENTS (SSNo, LastName, FirstName, HomePhone, Sex, DateofBirth, Age, Street, City, State, Zip)
    (ii) DOCTORS (SSNo, LastName, FirstName, OfficePhone, Pager, Specialty)
    (iii) BEDS (RoomNumber, BedNumber, Type, Status, PricePerHour)
    (iv) ACCOUNTS (DateIn, DateOut, Amount)

    Construct an E-R diagram for the hospital management; specify keys, mapping cardinalities, participation constraints (if necessary).

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A

(21119)

B.C.A.-V Sem.

Printed Pages : 3

Roll No. .....

18021

B.C.A. Examination, November–2019

INTRODUCTION TO DBMS

(BCA-501)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section–A

(Very Short Answer Questions)

Note : Attempt all the **five** questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 5×3=15

1. What do you mean by D.B.M.S. ?

2. Give example of a simple composite attributes of an entity.

3. What do you mean by referential integrity ?

4. What do you mean by DML and DDL ?

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5. Distinguish between functional dependency and multivalued dependency.

Section–B

(Short Answer Questions)

Note : Answer any ***two*** questions out of the following three questions. Each question carries 7½ marks. Short answer is required. 2×7½=15

6. What are the various anomalies associated with RDBMS.

7. What is union compatibility ? Explain.

8. What are the pit falls of lock based protocol ?

Section–C

(Detailed Answer Questions)

Note : Answer any ***three*** questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3×15=45

9. Define the normal forms. Explain the BCNF in detail.

10. Explain the superkey, primary key and candidate key with example in detail.

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11. What are the various characteristics of SQL ? Discuss five aggregate functions with suitable examples.

12. Discuss the selection, projection and join operator of relational algebra with suitable example.

13. What do you mean by Query and sub-query ? Discuss cursor in SQL also.

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(20221) Roll No. .....

BCA-V Sem.

18021

B.C.A. Examination, Dec.-2020

INTRODUCTION TO DBMS

(BCA-501)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section- A

(Very Short Answer Questions)

Note : Answer all the **five** questions. Each question carries 3 marks. Very short answer is required. 3×5=15

1. What are the advantages of file processing system which were removed by DBMS?

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2. Give example of a simple, composite attributes of an entity.
3. What do you mean by referential integrity?
4. What do you mean by DML and DDL?
5. What are anomalies in 1NF?

Section - B

Note : Answer any **Two** questions out of the following **Three** questions. Each question carries 7½ marks. Short answer is required. 7½×2=15

6. What do you mean by a key to retation? Explain the difference between super key, candidate key and primary key.
7. Define functional dependency. What do mean by loss-less decomposition?
8. Define Normal forms. List the definition of first, second and third normal forms.

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Section - C

Note : Answer any **Three** questions out of the following **five** questions. Each carries 15 marks. Answer is required in detail. 15×3=45

9. What is transactions? Draw a state diagram of transactions showing its states.

10. Explain ACID properties of a transaction with suitable example.

11. What are schedules? What are difference between conflict serializability and view serializability? https://www.ccsustudy.com

12. What are distributed database?

13. Describe the major problem associated with concurrent processing with example.

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D (Printed Pages 4)

(21223) Roll No. .....

B.C.A.-V Sem.

18021

B.C.A. Examination, Dec.-2023

Introduction to DBMS

(BCA-501)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt **all** questions. Each question carries **3** marks. Very short answer is required not exceeding 75 words.

3×5=15

1. Define data base management system?

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2. List any five application of D.B.M.S.
3. What are the disadvantages of file processing system?
4. Define Data model?
5. What are attributes? Give examples.

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions out of the following three questions. Each question carries 7.5 marks. Short answer is required not exceeding 200 words. 2×7.5=15

6. Compare generalization, specialization with example.
7. Explain the primary key, super key & foreign key.
8. What is normalization?

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Section-C

(Long Answer Type Questions)

Note : Attempt any **three** questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3×15=45

9. Draw an ER diagram for a small marketing company data-base, assuming your own data requirements.

10. Describe the classification of database language. Which type of language is SQL?

11. Explain the following :

    (i) DDL Command

    (ii) DML Command

12. Explain data independence with its types.

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13. Who are data administrators? What are the functions of database administrator?

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